Since their work, physicists have made steady progress in furthering the understanding of string cosmology, spearheaded by, among others, Gabriele Veneziano and his collaborator Maurizio Gasperini of the University of Torino. Gasperini and Veneziano have, in fact, come up with their own intriguing version of string cosmology that shares certain features with the scenario described above, but also differs in significant ways. As in the Brandenberger and Vafa work, they too rely on string theory's having a minimal length in order to avoid the infinite temperature and energy density that arises in the standard and inflationary cosmological theories. But rather than concluding that this means the universe begins as an extremely hot Planck-size nugget, Gasperini and Veneziano suggest that there may be a whole prehistory to the universe—starting long before what we have so far been calling time zero—that leads up to the Planckian cosmic embryo.

In this so-called pre-big bang scenario, the universe began in a vastly different state than it does in the big bang framework. Gasperini and Veneziano's work suggests that rather than being enormously hot and tightly curled into a tiny spatial speck, the universe started out as cold and essentially infinite in spatial extent. The equations of string theory then indicate that—somewhat as in Guth's inflationary epoch—an instability kicked in, driving every point in the universe to rush rapidly away from every other. Gasperini and Veneziano show that this caused space to become increasingly curved and results in a dramatic increase in temperature and energy density.7 After some time, a millimeter-sized three-dimensional region within this vast expanse could look just like the superhot and dense patch emerging from Guth's inflationary expansion. Then, through the standard expansion of ordinary big bang cosmology, this patch can account for the whole of the universe with which we are familiar. Moreover, because the pre-big bang epoch involves its own inflationary expansion, Guth's solution to the horizon problem is automatically built into the pre-big bang cosmological scenario. As Veneziano has said, "String theory offers us a version of inflationary cosmology on a silver platter."8

The study of superstring cosmology is rapidly becoming an active and fertile arena of research. The pre-big bang scenario, for example, has already generated a significant amount of heated, yet fruitful